Enriching data

DATA201/422 Data Wrangling

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What and why

The most important aspect of a statistical method: not what it does with the data but rather what data it uses.

Andrew Gelman's *Handy Statistical Lexicon*

- "Data enrichment" is enhancing a dataset by deriving new measures, or bringing in extra information
- Often the secret to a successful analysis is getting the right data
  - Not fitting the fanciest model

Plan for lectures

- Monday
  - What is enriching data?
  - Entity resolution
  - Adding extra variables
- Tuesday
  - Coverage and representativeness
  - Documentation
  - Role of AI

Entity resolution

The problem: Do these records refer to the same unit?

Different rows in same table

Address
3 Knuth Street
33 Knuth Street
Three Knuth St
3a Knuth Street
3 Knuth Road
3 Knuth St
3 Apple Street

Rows in different tables

Name	Income
Beth	38,800
Zoe	79,500
Ali	65,200

Name	Date of birth
Zoe	22-06-2001
Bethany	03-01-1997
Harry	30-05-1999
Alice	15-10-2005

Why is entity resolution important?

- Establishing internal structure of a dataset
  - Which hospital admissions belong to the same patient?
  - Which purchases were made by the same customer?
- Allowing linking between datasets
  - Bring together customers' in-store purchases and online purchases
- Importance often under-appreciated
  - E.g. the main purpose of loyalty cards is entity resolution

The Fellegi–Sunter method

- **m-probability** Probability that values agree, given that the records refer to the **same** entity
- **u-probability** Probability that values agree, given that the records refer to **different** entities
- Match likely when ratio of m-probability to u-probability is high

Data on Canterbury residents:

Value	m-probability	u-probability	ratio
Address: 20 Kirkwood Ave, Ilam	High	Low	High
Address: Christchurch	High	High	Moderate
Weight: 72.36 kg	Low	Low	Moderate

- For open-source software that implements this, see Splink

Entity resolution is often surprisingly hard

- Lots of data is still messy, inconsistent
- Can be slow on big datasets
  - Though “blocking” can help
- Costs of errors vary
  - Ad placement: low cost
  - Medical records used in treatments: high cost
  - Population estimation: high cost

Adding extra variables

Example

Data

products

id	name
1	Apple
2	Bread
3	Pear
4	Milk
5	Eggs

ratings

id	rating
3	yuck
1	yum
4	yum
5	yum
2	yuck

Results

id	name	rating
3	Pear	yuck
1	Apple	yum
4	Milk	yum
5	Eggs	yum
2	Bread	yuck

Example: The Integrated Data Infrastructure (IDI)

Contents:

- Health - e.g. pharmaceuticals, hospitalisations
- Education and training - e.g. school enrolment, industry training
- Benefits and social services data – e.g., benefits, student loans
- Justice - e.g. prisons, police records
- People and communities – e.g., driver licences, social surveys
- Population – e.g., border crossings, births, deaths, marriages
- Income and work – e.g. tax, Household Labour Force Survey
- Housing – e.g. tenancy, social housing

To integrate datasets, Stats NZ first links information from surveys, government agencies, and non-government organisations (NGOs) together using identifiable data, including first and last name, date of birth, age, sex, and country of birth.

Combining datasets with different classifications

- Often different datasets use different classifications
  - Boundaries for geographical units
  - Classification of occupations
  - Classification of diseases

Example: Constructing time series of population by area

Population in 2025

area	population
101	65
102	80
103	55

Population in 2026

area	population
101	70
102	85
104	20
105	40

Mappings between classifications

- Many different terms: Concordances, crosswalks, lookup tables, etc
- If organisations change their classification, they typically publish some sort of mapping between classifications

Example

area_2025	area_2026
101	101
102	102
103	104
103	105

Harmonising is sometimes easy, sometimes hard

Aggregating to coarser classification

Population

area	2025	2026
101	65	70
102	80	85
103	55	60

Splitting to finer classification

Population

area	2025	2026
101	65	70
102	80	85
104	?	20
105	?	40

Splitting methods

- Make some sort of simplifying assumption
  - Equal growth rates
  - Population distributed evenly across land area
  - Distribution for one group matches distribution for another group
- Requires subject matter knowledge
- Needs to be documented!

ASIDE

Data wrangling vs modelling

- Some data enrichment is deterministic
  - Straightforward inference that all analysts would agree on
  - E.g. converting “13 September 2026” to “2026-09-13”
- Some data enrichment is model-based or probabilistic
  - Incomplete information; different analysts take different approaches
  - E.g. assigning a birth date when all you know is the year of birth
- Place in the data science pipeline
  - Deterministic = “data wrangling”, early in pipeline
  - Model-based or probabilistic = “modelling”, later in pipeline
  - OK to sneak some model-based or probabilistic steps into data wrangling if the effect is tiny
  - Requires judgment calls (which need to be documented)

Missing values

- Adding extra variables often results in missing values
  - Coverage of datasets different
  - Extra data may be lower quality
- In general, don’t implement your own special method
  - R and Python have many sophisticated packages: use those
- Look for methods that can form plausible “Missing at Random” groups
  - More variables, more flexibility, typically helps
  - Machine learning models tend to be good (especially if dataset large)
- If you are including uncertainty measures in your analysis, look for “multiple imputation”